Categories, Proofs, and Processes:
Lecture 13 Yoneda's Lemma

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5 Yoneda's Lemma

5.1 Motivation

There are lots of Hom-functors, one for every $A \in \mathrm{Ob}(\mathcal{C})$.

... but they are all related by natural transformations.

$$t : \mathcal{C}(-, A) \to \mathcal{C}(-, B) := \{t_C : \mathcal{C}(C, A) \to \mathcal{C}(C, B)\}$$

For $f : C \to D$,

$$\begin{tikzcd}
\mathcal{C}(D, A) \arrow[r, "\mathcal{C}(f,A)"] \arrow[d, "t_D"'] & \mathcal{C}(C, A) \arrow[d, "t_C"] \\
\mathcal{C}(D, B) \arrow[r, "\mathcal{C}(f,B)"'] & \mathcal{C}(D, B)
\end{tikzcd}$$

The contravariant Hom-functor fixes the first element, the natural transfomration fixes the second element.

Claim 5.1 (Ex. 55). *For $g : A \to B$,*

$$t := \mathcal{C}(-, g) := \{t_C = \mathcal{C}(C, g) : \mathcal{C}(C, A) \to \mathcal{C}(C, B)\}$$

$\mathcal{C}(-, g)$ is a natural transformation.

Proof.

$$\begin{tikzcd}
\mathcal{C}(D, A) \arrow[rrr, "\mathcal{C}(f,A)"] \arrow[ddd, "t_D"'] & & & \mathcal{C}(C, A) \arrow[ddd, "t_C"] \\
& h \arrow[r, mapsto] \arrow[d, mapsto] & h \circ f \arrow[d, mapsto] & \\
& g \circ h \arrow[r, mapsto] & g \circ h \circ f & \\
\mathcal{C}(D, B) \arrow[rrr, "\mathcal{C}(f,B)"'] & & & \mathcal{C}(D, B)
\end{tikzcd}$$

□

Lemma 5.2 (Faux-neda)**.** *There exists a bijection*

$$\phi : Nat(\mathcal{C}(-, A), \mathcal{C}(-, B)) \cong \mathcal{C}(A, B)$$

5.2 Yoneda's Lemma

Lemma 5.3 (Yoneda's Lemma)**.** *For any functor $H : \mathcal{C}^{op} \to$ **Set**, and $A \in Ob(\mathcal{C})$, there exists a bijection (imagine $H = \mathcal{C}(-, B)$)*

$$\phi : Nat(\mathcal{C}(-, A), H) \cong H\ A$$

Proof. Take a natural transformation $t : \mathcal{C}(-, A) \to H$. We need an element $u \in H\ A$.

$$t_A : \mathcal{C}(A, A) \to H\ A$$

Let

$$u := t_A(1_A) \in H\ A, \quad \phi(t) = t_A(1_A)$$

For the inverse, take some $u \in H\ A$, and build a natural transformation t from it.

$$t_C : \mathcal{C}(C, A) \to H\ C$$

$$t_C(g) := H\ g\ u$$

Need to show that t is natural. $\forall f : C \to D$,

$$\begin{tikzcd}
\mathcal{C}(D, A) \arrow[rrr, "\mathcal{C}(f,A)"] \arrow[ddd, "t_D"'] & & & \mathcal{C}(C, A) \arrow[ddd, "t_C"] \\
& g \arrow[r, mapsto] \arrow[d, mapsto] & g \circ f \arrow[d, mapsto] & \\
& H\ g\ u \arrow[r, mapsto] & Hf(H\ g\ u) & \\
H\ D \arrow[rrr, "H\ f"'] & & & H\ C
\end{tikzcd}$$

$$Hf(H\ g\ u) = (Hf \circ Hg)u = H(g \circ f)u$$

since H is contravariant.

$$\phi(t) = t_A(1_A) \in H\ A$$

$$\psi(u) = t, \quad t_C(g) = H\ g\ u$$

$$\phi(\psi(u)) = \phi(t) = t_A(1_A) = H\ 1_A\ u = 1_{H\ A}\ u = u$$

$$\psi(\phi(s)) = \psi(s_A(1_A)) = t, \quad t_C(g) = (H\ g)(s_A(1_A))$$

$$\begin{aligned} t_C(g) &= (H\ g \circ s_A)\ 1_A \\ &= s_C\ \mathcal{C}(g, A)\ 1_A \\ &= s_C(g) \end{aligned} \tag{1}$$

For $g : C \to A$,

$$\begin{tikzcd} \mathcal{C}(A, A) \arrow[r, "\mathcal{C}(g,A)"] \arrow[d, "s_A"'] & \mathcal{C}(C, A) \arrow[d, "c_A"] \\ H\ A \arrow[r, "H\ g"'] & H\ C \end{tikzcd}$$

□

Corollary 5.4. *Yoneda's Lemma [5.3]* $\implies$ *Faux-neda [5.2].*

Example 5.5.

$$\phi : \mathrm{Nat}(\mathcal{C}(-, A), H) \cong H\ A$$

If $H = \mathcal{C}(-, B)$, then $\psi(f) : \mathcal{C}(-, A) \to \mathcal{C}(-, B)$,

$$\psi(f)_C(g) = \mathcal{C}(g, B)(f) = f \circ g$$

so $\psi(f) = \mathcal{C}(A, f)$

$$\phi(t) = \mathcal{C}(A, f)(1_A) = f$$

Fact 5.6. Yoneda embedding preserves all limits.